

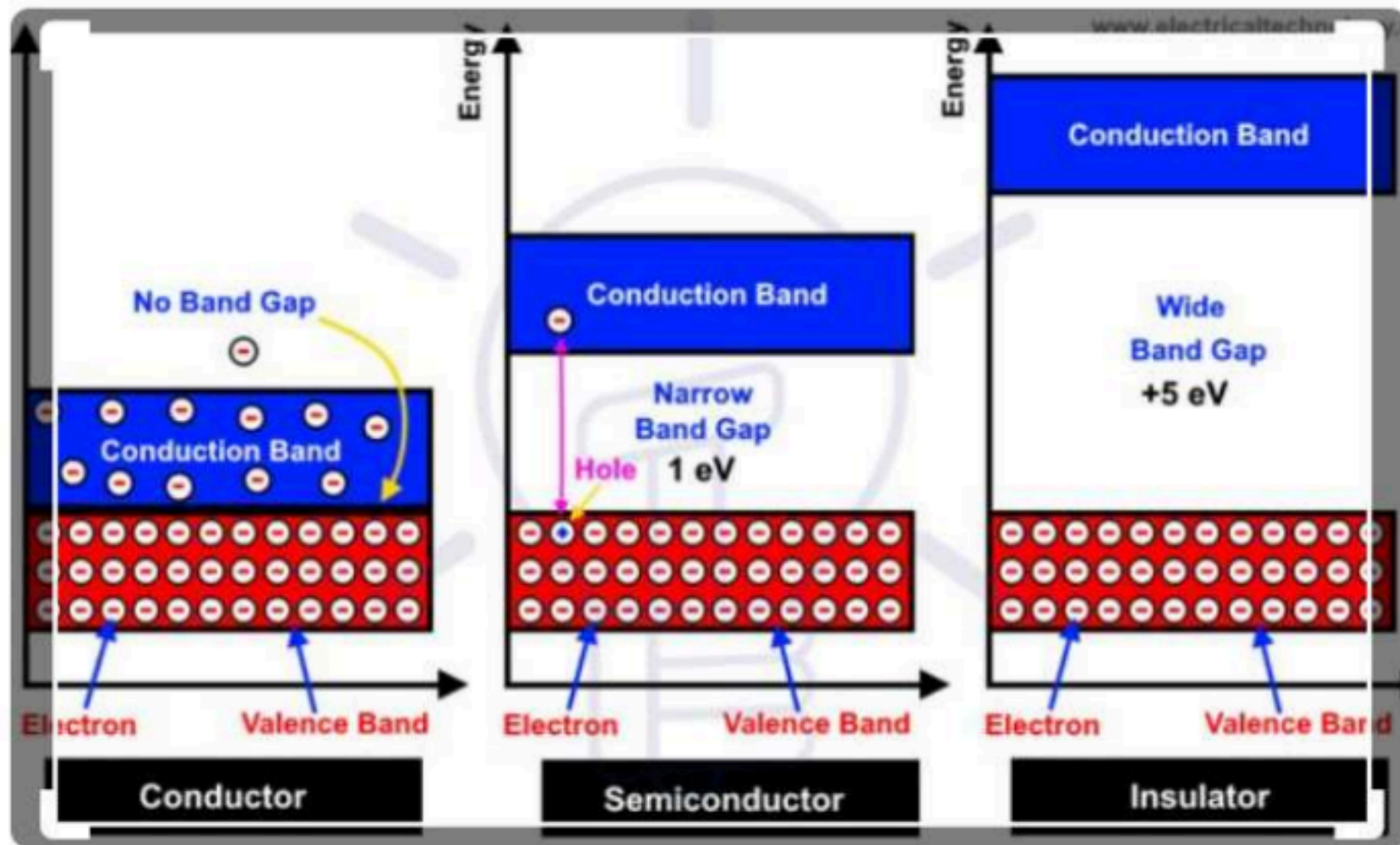
Chapter 1

Semiconductor Diode

(6hr)

Introduction to Semiconductors

- **Semiconductor** are materials whose electrical conductivity lies between **conductor** and an **insulator**.
 - Electrical Conductivity: Level to which a material conducts electricity.
 - **Conductor**: **allows** the current to flow through it with the application of voltage like **copper**.
 - **Insulator**: **Do not allows** the current to flow through it with the application of voltage like **Glass**.



Properties of Semiconductor

- It has negative temperature coefficient of resistance, i.e., the resistance of semiconductor decreases with increase in temperature and vice versa
- The resistivity lies between insulator and conductor
- Doping increases conductivity of semiconductor
- At absolute zero, it behaves as perfect insulator
- At room temperature it behaves as conductor.

Electron Current

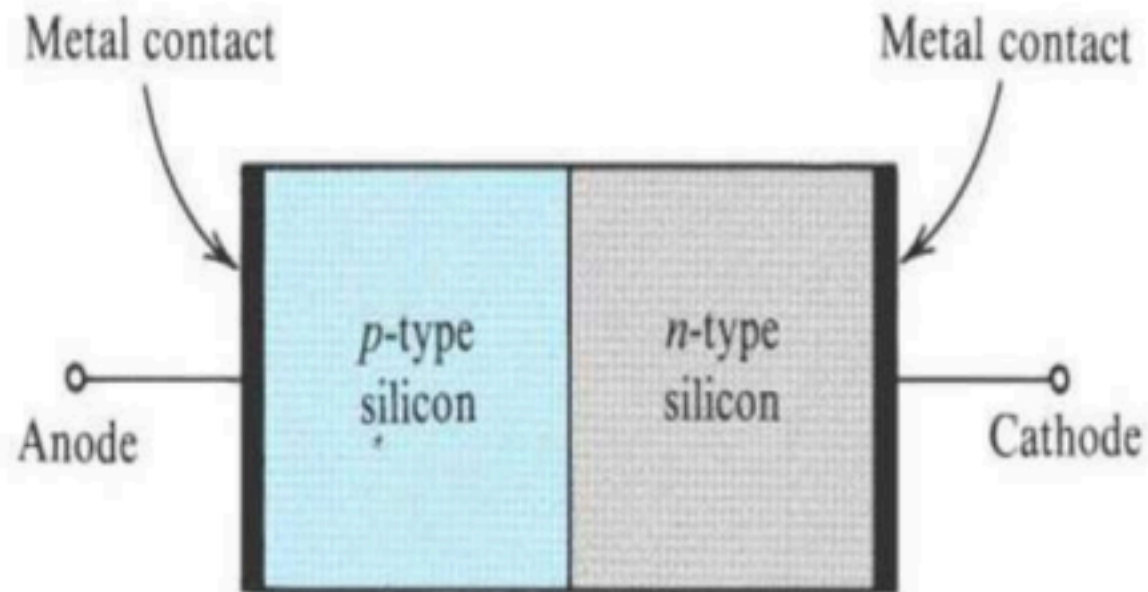
- ✚ At the temperature room, at any instant, a number of free electrons that are unattached to any atom drift randomly throughout the material. This condition occurs when no voltage is applied across a piece of intrinsic Si (as illustrated in Fig. 12).
- ✚ When a voltage is applied across the piece of intrinsic Si, as shown in Fig. 13, thermally generated free electrons in the conduction band, which are free to move, are now easily attracted toward the positive end.
- ✚ The movement of free electrons in a semiconductive material is called *electron current*.

- ✗ V_B for Ge is 0.3eV and 0.7eV for Si
- ✗ Barrier voltage depends on temperature
- ✗ V_B for both Ge and Si decreases by about 2 mV/ $^{\circ}\text{C}$
Therefore $\Delta V_B = -0.002 \Delta t$
where Δt is the rise in temperature
- ✗ V_B causes the drift of carriers through depletion layer. Hence barrier potential causes the drift current which is equal and opposite to diffusion current when final equilibrium is reached- Net current through the crystal is zero

Introduction: Diode

- The diode is the simplest and most fundamental nonlinear circuit element.
- Just like resistor, it has two terminals.
- Unlike resistor, it has a nonlinear current-voltage characteristics.
- Its use in rectifiers is the most common application.

Physical Structure of Diode



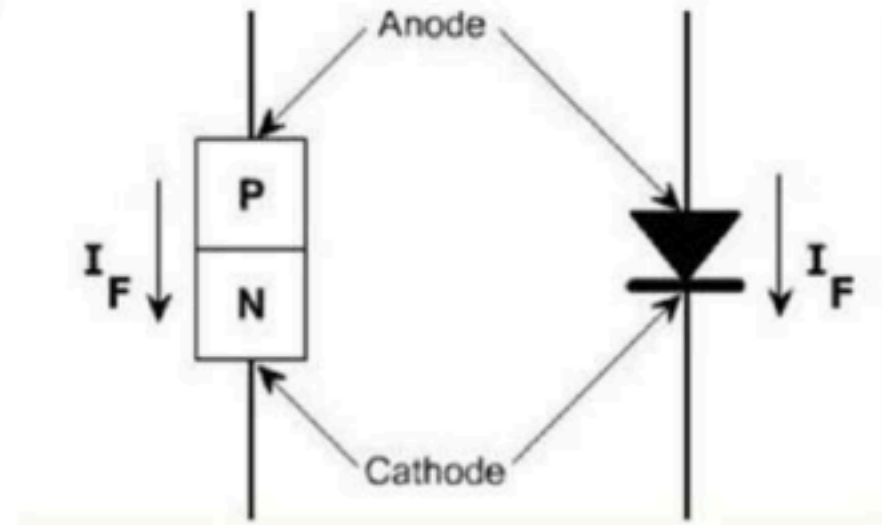
Diodes are formed by joining the N-type and P-type semiconductors

Diode is an electronic device which allows current to flow only in one direction

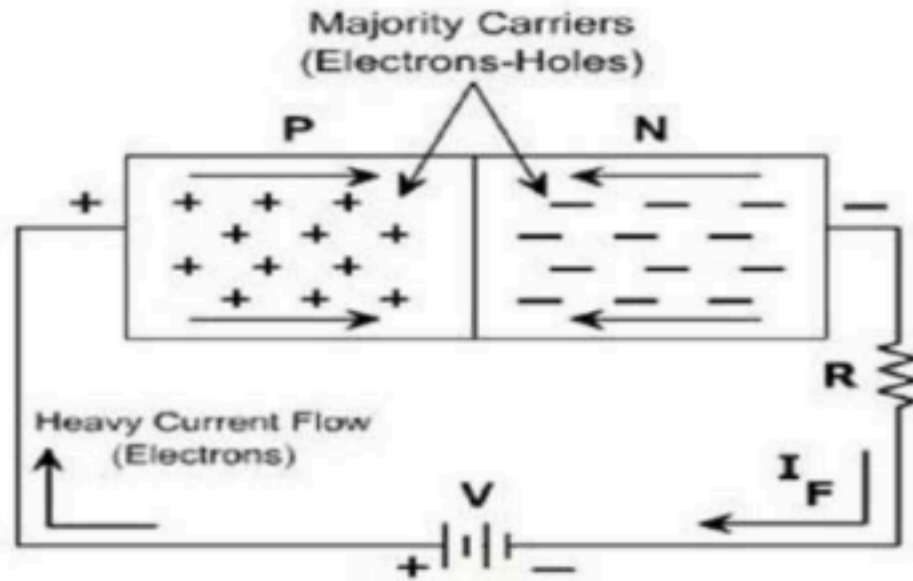
The voltage applied to the semiconductor diode is referred to as bias voltage.

Types of bias voltages:

- Forward Bias
- Reverse Bias

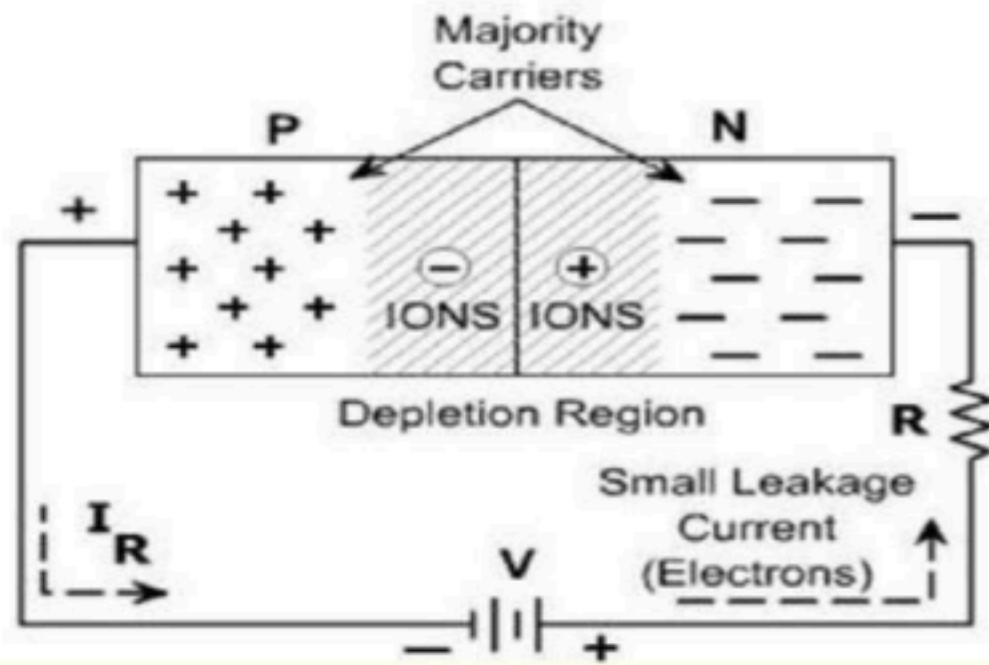


Forward Bias



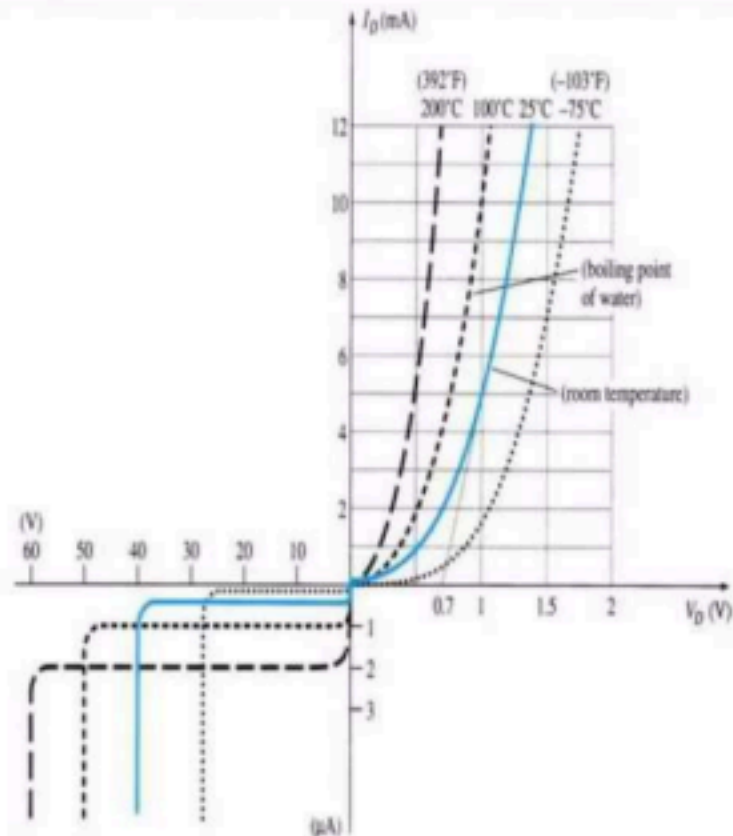
- A forward biased diode conducts current in the forward direction
- A forward biased diode conducts current as long as a high external bias voltage with correct polarity is applied

Reverse Bias



Extremely small current flows through the diode during reverse biasing. The minority carriers are forced towards the junction where they combine and support an extremely small current

Temperature Effects



Variation in Si diode characteristics with temperature change

- In the forward bias region the characteristics of a Si diode shift to the left at a rate of 2.5 mv per centigrade degree increase in temperature
- The reverse saturation current I_s will just about double in magnitude for every 10°C increase in temperature.
- The reverse breakdown voltage of a semiconductor diode will increase or decrease with temperature depending on the zener potential
- At room temperature Si and GaAs have relatively small reverse saturation current

$$I_D = I_S \left(e^{\frac{qV_D}{nkT}} - 1 \right)$$

Ideal Diode Equation

Where

I_D and V_D are the diode current and voltage, respectively

q is the charge on the electron

n is the ideality factor: $n = 1$ for indirect semiconductors (Si, Ge, etc.)

$n = 2$ for direct semiconductors (GaAs, InP, etc.)

k is Boltzmann's constant

T is temperature in Kelvin

kT/q is also known as V_{th} , the thermal voltage. At 300K (room temperature),

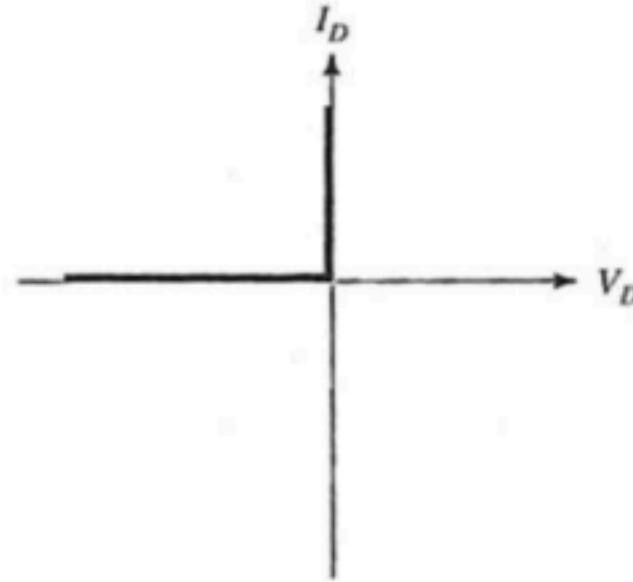
$kT/q = 25.9\text{mV}$

$$I_D = I_S(e^{V_D/\eta V_T} - 1)$$

Ideal and piecewise linear model of diode

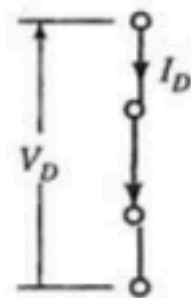
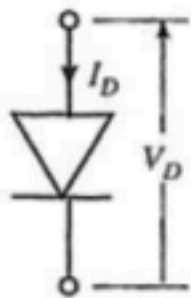
Ideal Diode

- ◆ Idealized two terminal device which passes current in one direction (zero resistance) and passes no current in the opposite direction (infinite resistance).
- ◆ Its v-i plot, which shows the relationship of the voltage across the diode and the current flowing through it, contains a discontinuity.

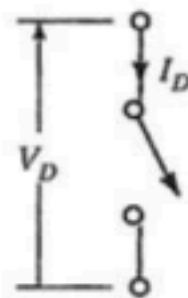


Ideal Diode

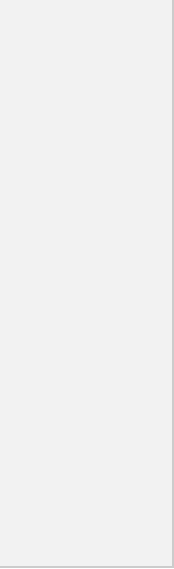
- ◆ If the diode is forward biased then the ideal diode conducts current as a closed switch.
- ◆ If the diode is reverse biased then the ideal diode will not conduct current, and it will appear as an open switch.



Forward bias
 $I > 0$
 $V = 0$
(closed switch)



Reverse bias
 $V < 0$
 $I = 0$
(open switch)

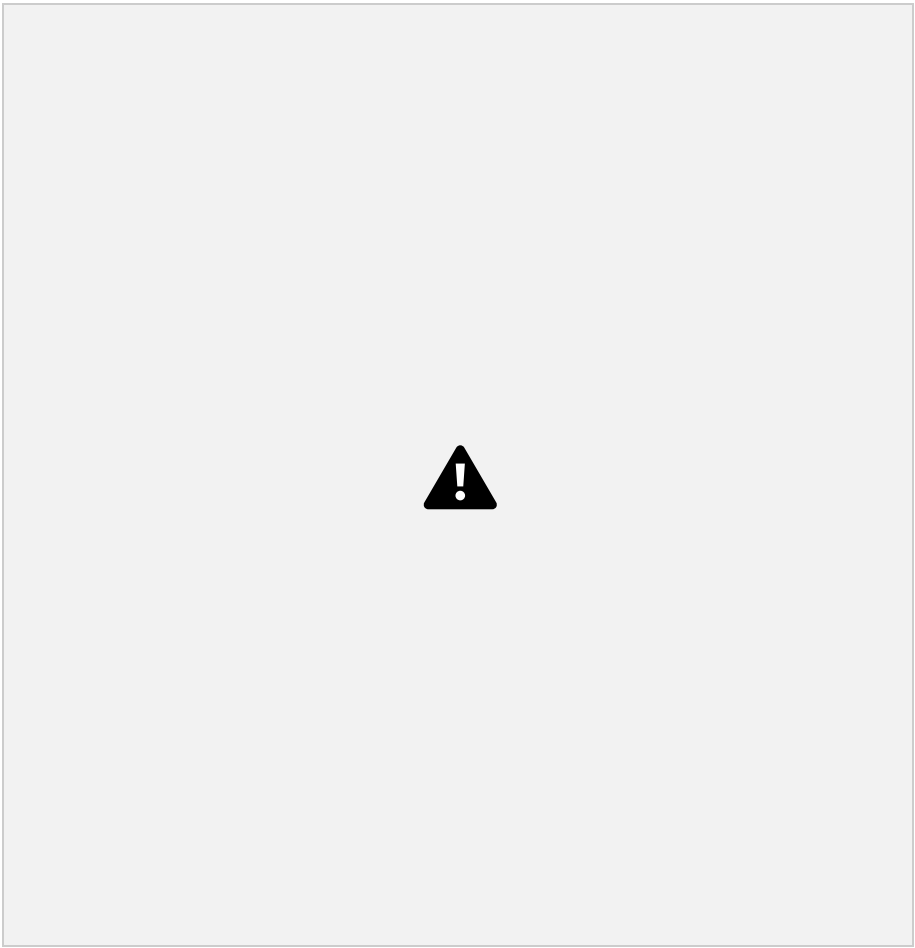

$$\begin{aligned}\text{Slope} &= \\ \frac{dy}{dx} &= \frac{1}{r_d} \\ &= I_D/V_D\end{aligned}$$







Formula of AC resistance or Dynamic Resistance of Diode



We have,



of diode r_d .

On solving we get dynamic resistance



Where value of Thermionic emission is 1 or 2 .

































